

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of MSc (Biochemistry) Examination May, 2018

BC855: Animal Cell Culture and Technology

Date: 04/05/2018 Day: Friday Time: 10.00 a.m. To 1.00 p.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	20
1.	One of the following is an antifungal agent (i) MRA (ii) Roccall (iii) tetracycline (iv) None of these	
2.	Which of the following cells would you consider as differentiated? (i) Blastomere (ii) muscle cells (iii) stem cells (iv) None of these	
3.	What is the major concern of using retroviral vectors to deliver or activate pluripotent genes? (i) not very efficient and needs repeated transfections (ii) slow and unstable (iii) vector integrates but is excised by the transposase (iv) vector and transgene remains in the genome and can be reactivated in differentiated cells	

4.	Dolly sheep was the first mammal to be cloned from an adult somatic cell. ☐True ☐False	
5.	Phenol red acts as indicator and turns tissue culture medium yellow at alkaline pH. ☐True ☐False	
6.	FCS is _____ serum. (i) foetal calf (ii) fibroblast containing (iii) foetal composed (iv) None of these	
7.	Cells such as macrophages and neurons can be used as secondary cultures since they do not divide <i>in vitro</i> . ☐True ☐False	
8.	Embryonic stem cells are generally isolated from inner cell mass of ____ (i) myoblast (ii) Blastocyst (iii) gastrula (iv) None of the above	
9.	The exact stage for cryofreezing during cryopreservation of a Finite cell line is around (i) 3rd generation (ii) 7 th generation (iii) 5 th generation (iv) None of the above	
10.	Serum doesn't (i) provide peptides that regulate growth and differentiation (ii) has essential nutrients such as vitamins, minerals etc (iii) provide fibronectin (iv) decrease viscosity of the medium	
11.	The most promising artificial blood product/s is/are (i) Perfluorocarbons (PFC s) (ii) Hemoglobin based oxygen carriers (HBOC) (iii) neither (i) nor (ii) (iv) both (i) and (ii)	
12.	Immune panning is the term referred to the attachment of cells to dishes coated with Abs ☐True ☐False	
13.	One of the following is not a method for testing cell viability (i) Clonogenic assay (ii) Neutral Red assay (iii) Calcein assay (iv) None of these	

14.	The cell culture may be heavily contaminated even in presence of antibiotics without any visible changes. Such type of contamination is known as ____ contamination (i) Biological (ii) physical (iii) cryptic (iv) chemical	
15.	Trypsinization is the process of cell dissociation that uses ____ enzyme. (i) lipolytic (ii) proteolytic (iii) amylolytic (iv) none of these	
16.	After a certain growth period when the cells reach confluency, the growth slows down and ceases finally. ☐True ☐False	
17.	Treating cells with colcemid before electroporation decreases the frequency of transfection ☐True ☐False	
18.	One of the following non-viral methods of gene delivery utilizes direct intracellular delivery of any water-soluble compounds and particles into internal organs. i) hydrodynamic ii) gene gun iii) magnetofaction iv) none of these	
19.	Production of active pharmaceutical substances in genetically modified organisms is referred as (i) molecular pharming (ii) molecular farming (iii) both (i) and(ii) (iv) none of these	
20.	A genetically modified mouse in which researchers have inactivated, an existing gene by replacing it or disrupting it with an artificial piece of DNA (i) knockout mice (ii) supermouse (iii) nude mouse (iv) none of these	

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Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q - II Answer the following questions as directed		20
	Answer <u>any five</u> of the following:	
1.	Define the importance of serum in tissue culture media citing suitable examples. Contrast the advantages and drawbacks associated with the application of serum in tissue culture media.	
2.	What physicochemical properties should be taken into consideration during media preparation? Discuss in detail.	
3.	Enlist different cell culture contaminants. Briefly explain their methods of detection.	
4.	Give a detailed note on stem cell culture based vaccines.	
5.	What is a HeLa cell line? Write its importance in animal tissue culture. Enlist various characteristics of transformed cell lines.	
6.	Define 'transgenic technology'. Discuss the role of this technology in creating transgenic animals with mentioning their relevant applications.	
SECTION - II		
Q-III Answer the following questions as directed		30
	Answer <u>any six</u> of the following:	
1.	How do cell cytotoxicity/viability tests differ from cell survival assays? Explain in detail with the help of one example of each type.	

2.	Give a detailed account on the type of instruments used in animal tissue culture labs alongwith their special features.	
3.	What do you mean by the term ‘Artificial Blood’? Describe the features of an ideal artificial blood and the technology used for its production.	
4.	Explain the bases of cell separation. Describe FACS (fluorescent activated cell sorting) technique as a method of cell separation process.	
5.	Differentiate the natural tissue culture media from artificial media. Define their composition and particular function mentioning proper examples.	
6.	Why do cells need to be Cryopreserved? Explain the principle behind action of cryoprotecting agents (CPA) and several phases involved in cryopreservation method.	
7.	Enlist different objectives behind the gene transfer in animals. Write down the steps of gene transfer and discuss electroporation method of gene transfer in animals.	